

Quiz 1

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) Given two vectors $\mathbf{u} = 3\mathbf{i} - 2\mathbf{j} + 6\mathbf{k}$ and $\mathbf{v} = (4, 0, 3)$ in $\mathbb{R}^3$.

(a) Find the angle between the two vectors $\mathbf{u}$ and $\mathbf{v}$.

$$\|\mathbf{u}\| = \sqrt{3^2 + (-2)^2 + 6^2} = \sqrt{49} = 7$$

$$\|\mathbf{v}\| = \sqrt{4^2 + 0^2 + 3^2} = \sqrt{25} = 5$$

$$\mathbf{u} \cdot \mathbf{v} = (3, -2, 6) \cdot (4, 0, 3) = 3 \times 4 + (-2) \times 0 + 6 \times 3 = 30$$

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} = \frac{30}{7 \times 5} = \frac{30}{35} = \frac{6}{7} \Rightarrow \theta = \cos^{-1}\left(\frac{6}{7}\right)$$

(b) Find the vector projection of $\mathbf{v}$ onto $\mathbf{u}$, $\text{proj}_{\mathbf{u}} \mathbf{v}$.

$$\text{proj}_{\mathbf{u}} \mathbf{v} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{u} \cdot \mathbf{u}} \right) \mathbf{u}$$

$$= \left(\frac{30}{7^2} \right) (3, -2, 6) = \left(\frac{30}{49} \right) (3, -2, 6)$$

$$= \left(\frac{90}{49}, -\frac{60}{49}, \frac{180}{49} \right)$$

(c) Compute $\mathbf{u} \times \mathbf{v}$ and then find the area of the parallelogram determined by $\mathbf{u}$ and $\mathbf{v}$.

$$\begin{aligned} \mathbf{u} \times \mathbf{v} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -2 & 6 \\ 4 & 0 & 3 \end{vmatrix} = \hat{i} \begin{vmatrix} -2 & 6 \\ 0 & 3 \end{vmatrix} - \hat{j} \begin{vmatrix} 3 & 6 \\ 4 & 3 \end{vmatrix} + \hat{k} \begin{vmatrix} 3 & -2 \\ 4 & 0 \end{vmatrix} \\ &= -6\hat{i} + 15\hat{j} + 8\hat{k} \end{aligned}$$

$$\begin{aligned} \Rightarrow \text{Area of parallelogram} &= \|\mathbf{u} \times \mathbf{v}\| = \|(-6, 15, 8)\| \\ &= \sqrt{(-6)^2 + 15^2 + 8^2} = \sqrt{325} = \underline{5\sqrt{13}} \end{aligned}$$

Problem 2. (10 points)

(a) Find parametric equations for the line through the points $P(1, 2, 3)$ and $Q(6, 3, 5)$.

Line direction $V = \overrightarrow{PQ} = (6, 3, 5) - (1, 2, 3)$
 $= (5, 1, 2)$

Parametric equations

$$L(t) = (1, 2, 3) + t(5, 1, 2)$$
$$= (1+5t, 2+t, 3+2t), \quad -\infty < t < +\infty$$

(b) Find an equation for the plane that passes through $P(0, 2, -1)$ and is parallel to the plane $3x - 2y - z = 1$.

Parallel to the plane $3x - 2y - z = 1 \Rightarrow \vec{n} = (3, -2, -1)$

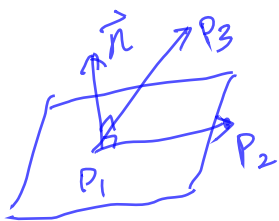
$\Rightarrow$ the equation of the plane

$$(3, -2, -1) \cdot (x-0, y-2, z+1) = 0$$

$$3x - 2(y-2) - (z+1) = 0$$

$$\underline{3x - 2y - z = -3}$$

(c) Find an equation for the plane that passes through the three points $P_1(0, 2, 1)$, $P_2(7, -1, 5)$, and $P_3(-1, 3, 0)$.



$$\overrightarrow{P_1P_2} = (7, -1, 5) - (0, 2, 1) = (7, -3, 4)$$

$$\overrightarrow{P_1P_3} = (-1, 3, 0) - (0, 2, 1) = (-1, 1, -1)$$

$$\overrightarrow{P_1P_2} \times \overrightarrow{P_1P_3} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 7 & -3 & 4 \\ -1 & 1 & -1 \end{vmatrix} = \vec{i} \begin{vmatrix} -3 & 4 \\ 1 & -1 \end{vmatrix} - \vec{j} \begin{vmatrix} 7 & 4 \\ -1 & -1 \end{vmatrix} + \vec{k} \begin{vmatrix} 7 & -3 \\ -1 & 1 \end{vmatrix}$$

$\Rightarrow$ the equation of the plane

$$(-1, 3, 4) \cdot (x-0, y-2, z-1) = 0$$

$$-(x) + 3(y-2) + 4(z-1) = 0$$

$$\underline{-x + 3y + 4z = 10}$$